Closed-Form Swaption Pricing and Calibration under the Gaussian G3++ / HJM Framework

with Application to USD SOFR Swaption Normal Volatility

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Abstract

We present a self-contained derivation of the closed-form swaption pricing formula under the three-factor Gaussian Heath–Jarrow–Morton (HJM) framework, known as the G3++ model. Starting from the fundamental HJM no-arbitrage drift condition, we derive the affine bond pricing formula, construct the forward swap rate as a ratio of affine functionals, and apply a first-order linearisation to obtain a Gaussian distribution for the forward swap rate at expiry. This yields a closed-form Bachelier (normal) implied volatility expression directly suitable for USD SOFR swaption markets. We then develop a rigorous two-stage calibration algorithm exploiting the convex structure of the problem in the volatility amplitudes, present analytical gradients for efficient optimisation, and discuss convergence criteria, parameter constraints, PCA-based initialisation, and practical diagnostics.

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1. Model Setup

1.1. Forward Rate Dynamics

We work on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{Q})$ where $\mathbb{Q}$ denotes the risk-neutral measure. Let $f(t,T)$ denote the instantaneous forward rate at time t for maturity $T \geq t$, and let

$$P(t,T) \;=\; \exp\left(-\int_t^T f(t,u)\,\mathrm{d}u\right)$$

denote the zero-coupon bond price. The risk-neutral bank account (money-market account) is defined as

$$B(0,t) \;:=\; e^{\int_0^t r(s)\,\mathrm{d}s}, \tag{1}$$

where $r(t) = f(t,t)$ is the short rate. The ratio $P(t,T)/B(0,t)$ is a $\mathbb{Q}$-martingale by the fundamental theorem of asset pricing, and $B(0,t)$ serves as the reference numeraire for

measure changes throughout this paper.

Under $\mathbb{Q}$, the instantaneous forward rate evolves according to the three-factor HJM stochastic differential equation:

$$\mathrm{d}f(t,T) \;=\; \alpha(t,T)\,\mathrm{d}t \;+\; \sum_{k=1}^{3} \sigma_k\, e^{-\kappa_k(T-t)}\,\mathrm{d}W_k(t), \tag{2}$$

where $\{W_1, W_2, W_3\}$ are correlated standard Brownian motions under $\mathbb{Q}$ satisfying

$$\mathrm{d}W_k(t)\,\mathrm{d}W_l(t) \;=\; \rho_{kl}\,\mathrm{d}t, \qquad k,l \in \{1,2,3\}. \tag{3}$$

The volatility function for factor k takes the separable exponential form

$$\sigma_k(t,T) \;=\; \sigma_k\, e^{-\kappa_k(T-t)}, \tag{4}$$

where $\sigma_k > 0$ is the volatility amplitude and $\kappa_k > 0$ is the mean-reversion (decay) rate. The three factors correspond to the dominant empirical shapes of the forward curve: level ($k = 1$, slow decay, near-flat across tenors), slope ($k = 2$, intermediate decay), and curvature ($k = 3$, fast decay, short-end dominant). PCA of historical SOFR forward rate changes confirms that these three components explain over 95% of total variance (Section 6.6); indicative variance contributions are:

Factor	**Shape**	**Variance explained**	**Cumulative**
$k = 1$	Level	∼85%	∼85%
$k = 2$	Slope	∼8%	∼93%
$k = 3$	Curvature	∼3%	∼96%

Based on USD SOFR daily forward rate changes over 2021–2025, spanning low-volatility (pre-tightening), high-volatility (2022–2023 FOMC cycle), and normalisation regimes. The three G3++ factors collectively capture ∼96% of total variance; the remaining ∼4% is diffuse across higher eigenvectors and offers diminishing calibration benefit relative to the added parameter complexity, justifying the three-factor truncation.

1.2. HJM No-Arbitrage Drift Condition

A fundamental result of Heath et al. (1992) states that under $\mathbb{Q}$, the drift $\alpha(t,T)$ is completely determined by the volatility functions and is *not* a free parameter. Specifically:

$$\alpha(t,T) \;=\; \sum_{k=1}^{3}\sum_{l=1}^{3} \rho_{kl}\,\sigma_k(t,T) \int_t^T \sigma_l(t,u)\,\mathrm{d}u. \tag{5}$$

Substituting (4) into (5), the inner integral evaluates to

$$\int_t^T \sigma_l(t,u)\,\mathrm{d}u \;=\; \frac{\sigma_l}{\kappa_l}\big(1 - e^{-\kappa_l(T-t)}\big) \;=:\; \sigma_l\, B_l(t,T), \tag{6}$$

where we define the fundamental mean-reversion integral

$$\boxed{B_k(t,T) \;:=\; \frac{1 - e^{-\kappa_k(T-t)}}{\kappa_k}.} \tag{7}$$

The function $B_k(t,T)$ satisfies $0 \le B_k(t,T) \le \kappa_k^{-1}$, with $B_k(t,T) \to T - t$ as $\kappa_k \to 0$ (parallel shift limit) and $B_k(t,T) \to \kappa_k^{-1}$ as $T - t \to \infty$.

1.3. State-Space (G3++) Representation

Define the three Gaussian state variables

$$x_k(t) \;:=\; \int_0^t \sigma_k\, e^{-\kappa_k(t-s)}\, \mathrm{d}W_k(s), \qquad k = 1, \ldots, 3. \tag{8}$$

Each x_k satisfies the Ornstein–Uhlenbeck SDE under $\mathbb{Q}$:

$$\mathrm{d}x_k(t) \;=\; -\kappa_k\, x_k(t)\, \mathrm{d}t \;+\; \sigma_k\, \mathrm{d}W_k(t), \qquad x_k(0) = 0. \tag{9}$$

The short rate decomposes as

$$r(t) \;=\; f(t,t) \;=\; \sum_{k=1}^{3} x_k(t) \;+\; \varphi(t), \tag{10}$$

where $\varphi(t)$ is a deterministic shift chosen to fit the initial discount curve $P(0,T)$ exactly. As shown in Section 2, the explicit form of $\varphi(t)$ is not required for pricing: it cancels analytically in all bond price ratios.

2. Closed-Form Zero-Coupon Bond Pricing

2.1. The Affine Bond Price Formula

Theorem 2.1 (Affine Bond Price)**.** *Under the G3++ dynamics* (2)–(9)*, the zero-coupon bond price at time t with maturity T is given by*

$$P(t,T) \;=\; \frac{P(0,T)}{P(0,t)} \exp\!\left(-\sum_{k=1}^{3} B_k(t,T)\, x_k(t) \;+\; \frac{1}{2} \sum_{k=1}^{3} \sum_{l=1}^{3} \rho_{kl}\, \sigma_k \sigma_l\, \Gamma_{kl}(t,T) \right), \tag{11}$$

where the variance integral $\Gamma_{kl}(t,T) := \int_t^T B_k(t,u)\, B_l(t,u)\, \mathrm{d}u$ has the closed form

$$\Gamma_{kl}(t,T) \;=\; \frac{1 - e^{-(\kappa_k+\kappa_l)(T-t)}}{\kappa_k \kappa_l (\kappa_k + \kappa_l)} \;-\; \frac{B_k(t,T)}{\kappa_l(\kappa_k+\kappa_l)} \;-\; \frac{B_l(t,T)}{\kappa_k(\kappa_k+\kappa_l)} \;+\; \frac{T-t}{\kappa_k \kappa_l}. \tag{12}$$

Proof. Since the volatility functions (4) are deterministic, $f(t,T)$ is Gaussian and bond prices are log-normal. Write $\log P(t,T) = -\int_t^T f(t,u)\, \mathrm{d}u$ and apply Itô's lemma to collect drift and quadratic variation terms. The ratio $P(0,T)/P(0,t)$ enforces exact calibration to the initial curve, which eliminates $\varphi(t)$ analytically. See Brigo and Mercurio (2006), Chapter 4, for the complete proof. □

> **Remark 2.1** (Initial Curve Fitting). The factor $P(0,T)/P(0,t)$ in (11) guarantees exact fit to any observed initial discount curve $P(0,T)$ without additional calibration.

3. Forward Swap Rate and Swaption Payoff

3.1. Forward Swap Rate

Consider a swaption with expiry T_e and swap tenor T_s. The underlying swap has payment dates $T_e + T_1, \ldots, T_e + T_n$, where $T_i = i\delta$, $T_n = T_s$, and δ is the accrual fraction ($\frac{1}{2}$ for semi-annual, $\frac{1}{4}$ for quarterly SOFR). Define the annuity

$$A(t; T_e, T_s) \;:=\; \sum_{i=1}^{n} \delta_i \, P(t, T_e + T_i), \tag{13}$$

and denote its time-0 value by $A_0 := A(0; T_e, T_s)$, which serves as the normalising constant for the annuity measure throughout this paper. The at-the-money forward swap rate — the fixed rate that makes the swap value zero — is

$$F(t; T_e, T_s) \;=\; \frac{P(t, T_e) - P(t, T_e + T_s)}{A(t; T_e, T_s)}. \tag{14}$$

Under the annuity (swap) measure $\mathbb{Q}^A$ with numeraire $A(t; T_e, T_s)$, the forward swap rate $F(t; T_e, T_s)$ is a $\mathbb{Q}^A$-martingale.

3.2. Swaption Payoff and Coupon Bond Representation

A payer swaption with strike K pays at expiry T_e:

$$V(T_e) \;=\; A(T_e; T_e, T_s) \cdot \big(F(T_e; T_e, T_s) - K\big)^+. \tag{15}$$

Using (14), the payoff rewrites as a put on a coupon bond:

$$V(T_e) \;=\; \left(1 - \sum_{i=1}^{n} c_i \, P(T_e, T_e + T_i)\right)^+, \tag{16}$$

where $c_i = K\delta_i$ for $i < n$ and $c_n = 1 + K\delta_n$. In a single-factor model, Jamshidian's decomposition (Jamshidian, 1989) prices (16) exactly as a sum of zero-coupon bond options. In the three-factor setting this exact decomposition is unavailable, and we instead apply the following linearisation.

4. Derivation of the Forward Swap Rate SDE

This section derives the exact stochastic differential equation satisfied by the forward swap rate $F(t; T_e, T_s)$ under both the risk-neutral measure $\mathbb{Q}$ and the annuity (swap) measure $\mathbb{Q}^A$. The derivation proceeds in six steps: (i) derive the bond price SDE from the G3++ dynamics; (ii) compute the annuity SDE; (iii) apply Itô's quotient rule to obtain the exact swap rate SDE under $\mathbb{Q}$; (iv) change measure to $\mathbb{Q}^A$ via Girsanov's theorem; (v) evaluate the instantaneous variance and introduce the frozen-coefficient approximation; (vi) identify the resulting volatility coefficients with the hedge ratios of Section 5.

4.1. Step 1 — Bond Price SDE Under $\mathbb{Q}$

Starting from the affine bond price formula (11), write

$$P(t,T) \;=\; G(t,T)\,\exp\Big(-\sum_{k=1}^{3} B_k(t,T)\,x_k(t)\Big),$$

where $G(t,T)$ collects all deterministic terms. Applying Itô's lemma and the OU dynamics $\mathrm{d}x_k = -\kappa_k x_k\,\mathrm{d}t + \sigma_k\,\mathrm{d}W_k$:

Proposition 4.1 (Bond Price SDE)**.** *Under the risk-neutral measure $\mathbb{Q}$,*

$$\frac{\mathrm{d}P(t,T)}{P(t,T)} \;=\; r(t)\,\mathrm{d}t \;-\; \sum_{k=1}^{3} \sigma_k\,B_k(t,T)\,\mathrm{d}W_k(t). \tag{17}$$

Proof. Differentiate $\log P = \log G - \sum_k B_k(t,T)\,x_k$ using Itô's formula. The time derivative of $B_k(t,T)$ satisfies

$$\frac{\partial B_k(t,T)}{\partial t} \;=\; -e^{-\kappa_k(T-t)} \;=\; \kappa_k\,B_k(t,T) - 1. \tag{18}$$

Substituting the OU dynamics and collecting the $\mathrm{d}t$ terms, one finds that the drift of $\mathrm{d}(\log P)$ equals $r(t)$, after the quadratic-variation term $\frac{1}{2}\sum_{k,l}\rho_{kl}\sigma_k\sigma_l B_k B_l\,\mathrm{d}t$ cancels with the HJM drift (5). The diffusion terms give $-\sum_k \sigma_k B_k(t,T)\,\mathrm{d}W_k$. Exponentiating yields (17). □

The factor $-\sigma_k B_k(t,T)$ is the *dollar duration* of the bond with respect to factor k: a unit positive shock in W_k reduces $P(t,T)$ by $\sigma_k B_k(t,T)$ in relative terms. The sensitivity $B_k(t,T) = (1-e^{-\kappa_k(T-t)})/\kappa_k$ increases with maturity and saturates at $1/\kappa_k$, so long-dated bonds are more sensitive to slow factors (small κ_k) and short-dated bonds to fast factors (large κ_k).

4.2. Step 2 — Annuity SDE Under $\mathbb{Q}$

Since the annuity $A(t;T_e,T_s) = \sum_{i=1}^{n}\delta_i P(t,T_e+T_i)$ is a positive linear combination of bond prices, Itô's lemma for sums gives

$$\frac{\mathrm{d}A(t)}{A(t)} \;=\; r(t)\,\mathrm{d}t \;-\; \sum_{k=1}^{3} \sigma_k\,\mathcal{B}_k^A(t)\,\mathrm{d}W_k(t), \tag{19}$$

where $\mathcal{B}_k^A(t)$ is the *annuity-weighted average duration* in factor k — distinct from the single-bond duration $B_k(t,T)$ of equation (7) — defined as

$$\mathcal{B}_k^A(t;T_e,T_s) \;:=\; \frac{\displaystyle\sum_{i=1}^{n}\delta_i\,B_k(t,T_e{+}T_i)\,P(t,T_e{+}T_i)}{A(t;T_e,T_s)}. \tag{20}$$

Equation (19) has the same form as (17): the annuity grows at $r(t)$ and carries an effective factor duration $\mathcal{B}_k^A(t)$, obtained by weighting each bond duration $B_k(t,T_e{+}T_i)$ by its present-value share $\delta_i P(t,T_e{+}T_i)/A(t)$ of the annuity.

4.3. Step 3 — Exact Swap Rate SDE Under $\mathbb{Q}$

Write $F(t) = N(t)/A(t)$ where the floating-leg numerator is $N(t) := P(t, T_e) - P(t, T_e + T_s)$. From (17), the numerator SDE is

$$\frac{\mathrm{d}N(t)}{N(t)} = r(t)\ \mathrm{d}t - \sum_{k=1}^{3} \sigma_k\, \mathcal{B}_k^N(t)\ \mathrm{d}W_k(t), \tag{21}$$

with the *numerator-weighted average duration* in factor k

$$\mathcal{B}_k^N(t; T_e, T_s) := \frac{B_k(t, T_e)\, P(t, T_e) - B_k(t, T_e{+}T_s)\, P(t, T_e{+}T_s)}{N(t)}. \tag{22}$$

For any Itô process X, we write $\mathrm{d}\langle X\rangle_t$ for its instantaneous quadratic variation and $\mathrm{d}\langle X, Y\rangle_t$ for the cross-variation of two processes X and Y, both defined via

$$\mathrm{d}\langle X\rangle_t := (\mathrm{d}X)^2, \qquad \mathrm{d}\langle X, Y\rangle_t := \mathrm{d}X \cdot \mathrm{d}Y, \tag{23}$$

where products of differentials are evaluated using the rules $\mathrm{d}W_k\, \mathrm{d}W_l = \rho_{kl}\, \mathrm{d}t$ and $(\mathrm{d}t)^2 = \mathrm{d}t\, \mathrm{d}W_k = 0$. Applying Itô's lemma $\mathrm{d}(\log X) = \mathrm{d}X/X - \frac{1}{2}\mathrm{d}\langle \log X\rangle$ to (21) and (19) gives the log differentials:

$$\mathrm{d}(\log N(t)) = \Big(r(t) - \tfrac{1}{2}\sum_{k,l} \rho_{kl}\sigma_k\sigma_l \mathcal{B}_k^N \mathcal{B}_l^N\Big)\, \mathrm{d}t - \sum_{k=1}^{3} \sigma_k\, \mathcal{B}_k^N(t)\ \mathrm{d}W_k(t), \tag{24}$$

$$\mathrm{d}(\log A(t)) = \Big(r(t) - \tfrac{1}{2}\sum_{k,l} \rho_{kl}\sigma_k\sigma_l \mathcal{B}_k^A \mathcal{B}_l^A\Big)\, \mathrm{d}t - \sum_{k=1}^{3} \sigma_k\, \mathcal{B}_k^A(t)\ \mathrm{d}W_k(t). \tag{25}$$

The quadratic variations and cross-variation then follow immediately from squaring and cross-multiplying the diffusion terms:

$$\mathrm{d}\langle \log N\rangle_t = \sum_{k=1}^{3}\sum_{l=1}^{3} \rho_{kl}\, \sigma_k\sigma_l\, \mathcal{B}_k^N(t)\, \mathcal{B}_l^N(t)\ \mathrm{d}t, \tag{26}$$

$$\mathrm{d}\langle \log A\rangle_t = \sum_{k=1}^{3}\sum_{l=1}^{3} \rho_{kl}\, \sigma_k\sigma_l\, \mathcal{B}_k^A(t)\, \mathcal{B}_l^A(t)\ \mathrm{d}t, \tag{27}$$

$$\mathrm{d}\langle \log N, \log A\rangle_t = \sum_{k=1}^{3}\sum_{l=1}^{3} \rho_{kl}\, \sigma_k\sigma_l\, \mathcal{B}_k^N(t)\, \mathcal{B}_l^A(t)\ \mathrm{d}t. \tag{28}$$

Since $\log(N/A) = \log N - \log A$, the quadratic variation of the log-swap-rate is

$$\begin{aligned} \mathrm{d}\Big\langle \log \frac{N}{A}\Big\rangle_t &= \mathrm{d}\langle \log N\rangle_t - 2\ \mathrm{d}\langle \log N, \log A\rangle_t + \mathrm{d}\langle \log A\rangle_t \\ &= \sum_{k,l} \rho_{kl}\, \sigma_k\sigma_l \big(\mathcal{B}_k^N - \mathcal{B}_k^A\big)\big(\mathcal{B}_l^N - \mathcal{B}_l^A\big)\ \mathrm{d}t. \end{aligned} \tag{29}$$

Applying Itô's quotient rule $\mathrm{d}F = F\ \mathrm{d}(\log F) + \frac{1}{2}F\ \mathrm{d}\langle \log F\rangle$ to $F = N/A$:

$$\begin{aligned} \mathrm{d}F &= F\ \mathrm{d}\Big(\log \frac{N}{A}\Big) + \tfrac{1}{2}F\ \mathrm{d}\Big\langle \log \frac{N}{A}\Big\rangle \\ &= F\Big[\big(\tfrac{\mathrm{d}N}{N} - \tfrac{\mathrm{d}A}{A}\big) + \underbrace{\mathrm{d}\langle \log A\rangle - \mathrm{d}\langle \log N, \log A\rangle}_{\text{Itô correction}}\Big]. \end{aligned} \tag{30}$$

Substituting (19) and (21), the $r(t)\ \mathrm{d}t$ terms cancel, and we define the *net factor volatility* of the swap rate:

$$\gamma_k(t; T_e, T_s) \ := \ \sigma_k\big(\mathcal{B}_k^A(t) - \mathcal{B}_k^N(t)\big). \tag{31}$$

The Itô correction in (30) evaluates to $\sum_{k,l} \rho_{kl}\sigma_k\sigma_l\mathcal{B}_k^A(\mathcal{B}_l^A - \mathcal{B}_l^N)\ \mathrm{d}t$.

Proposition 4.2 (Exact Swap Rate SDE Under $\mathbb{Q}$)**.**

$$\frac{\mathrm{d}F(t)}{F(t)} \ = \ \mu_F(t)\ \mathrm{d}t \ + \ \sum_{k=1}^{3} \gamma_k(t)\ \mathrm{d}W_k(t), \tag{32}$$

where the risk-neutral drift is

$$\mu_F(t) \ = \ \sum_{k=1}^{3}\sum_{l=1}^{3} \rho_{kl}\,\sigma_k\sigma_l\,\mathcal{B}_k^A(t)\big(\mathcal{B}_l^A(t) - \mathcal{B}_l^N(t)\big). \tag{33}$$

Remark. The net factor volatility $\gamma_k = \sigma_k(\mathcal{B}_k^A - \mathcal{B}_k^N)$ measures how much the annuity and the floating-leg numerator *differ* in their sensitivity to factor k. Economically, F is the ratio N/A, so its volatility arises from imperfect correlation between numerator and denominator: if both moved identically ($\mathcal{B}_k^N = \mathcal{B}_k^A$), F would be constant. The annuity has contributions from all cashflow dates $T_e + T_1, \ldots, T_e + T_n$, while the numerator only depends on the two endpoint bonds $P(\cdot, T_e)$ and $P(\cdot, T_e{+}T_s)$. This asymmetry in tenor profile drives the non-zero γ_k.

4.4. Step 4 — Change of Measure to $\mathbb{Q}^A$ via Girsanov's Theorem

The annuity $A(t; T_e, T_s)$ is the natural numeraire for swaption pricing. Under the annuity measure $\mathbb{Q}^A$, the Radon–Nikodým density on $\mathcal{F}_t$ is

$$\left.\frac{\mathrm{d}\mathbb{Q}^A}{\mathrm{d}\mathbb{Q}}\right|_{\mathcal{F}_t} \ = \ \frac{A(t)/A_0}{B(0,t)}, \qquad B(0,t) := e^{\int_0^t r(s)\ \mathrm{d}s}. \tag{34}$$

The Radon–Nikodým density (34) defines the annuity measure via the stochastic exponential of the process $\log(A(t)/A_0) - \log B(0,t)$. By Girsanov's theorem for *correlated* Brownian motions, if the log-numeraire has diffusion $\sum_l \theta_l\ \mathrm{d}W_l$ then the drift added to $\mathrm{d}W_k$ under the new measure is $\sum_l \rho_{kl}\,\theta_l\ \mathrm{d}t$ — the full row k of the correlation matrix acting on the Girsanov kernel vector. From (25), the diffusion of $\log A$ has kernel $\theta_l = -\sigma_l\mathcal{B}_l^A$, so:

$$\mathrm{d}W_k^A(t) \ = \ \mathrm{d}W_k(t) \ + \ \sum_{l=1}^{3} \rho_{kl}\,\sigma_l\,\mathcal{B}_l^A(t)\ \mathrm{d}t, \qquad k = 1, \ldots, 3. \tag{35}$$

The $\mathbb{Q}^A$-Brownian motions remain correlated: $\mathrm{d}W_k^A\ \mathrm{d}W_l^A = \rho_{kl}\ \mathrm{d}t$.

Substituting (35) into (32), each $\mathrm{d}W_k = \mathrm{d}W_k^A - \sum_l \rho_{kl}\sigma_l\mathcal{B}_l^A\ \mathrm{d}t$. The diffusion $\sum_k \gamma_k\ \mathrm{d}W_k$ becomes

$$\sum_{k=1}^{3} \gamma_k\ \mathrm{d}W_k \ = \ \sum_{k=1}^{3} \gamma_k\ \mathrm{d}W_k^A \ - \ \underbrace{\sum_{k=1}^{3}\sum_{l=1}^{3} \rho_{kl}\,\gamma_k\,\sigma_l\,\mathcal{B}_l^A}_{\text{drift correction}}\ \mathrm{d}t. \tag{36}$$

Substituting $\gamma_k = \sigma_k(\mathcal{B}_k^A - \mathcal{B}_k^N)$, the total drift of $\mathrm{d}F/F$ under $\mathbb{Q}^A$ is

$$\mu_F^{\mathbb{Q}^A} \;=\; \mu_F \;-\; \sum_{k,l} \rho_{kl}\, \sigma_k \sigma_l \big(\mathcal{B}_k^A - \mathcal{B}_k^N\big)\mathcal{B}_l^A. \tag{37}$$

Recalling (33), $\mu_F = \sum_{k,l} \rho_{kl}\sigma_k\sigma_l\mathcal{B}_k^A(\mathcal{B}_l^A - \mathcal{B}_l^N)$. Relabelling the dummy summation indices $k \leftrightarrow l$ in the correction term and using symmetry $\rho_{lk} = \rho_{kl}$:

$$\begin{aligned}\sum_{k,l} \rho_{kl}\sigma_k\sigma_l\big(\mathcal{B}_k^A - \mathcal{B}_k^N\big)\mathcal{B}_l^A \;&=\; \sum_{k,l} \rho_{lk}\sigma_l\sigma_k\big(\mathcal{B}_l^A - \mathcal{B}_l^N\big)\mathcal{B}_k^A \\ &=\; \sum_{k,l} \rho_{kl}\sigma_k\sigma_l\, \mathcal{B}_k^A\big(\mathcal{B}_l^A - \mathcal{B}_l^N\big) \;=\; \mu_F.\end{aligned}$$

Therefore $\mu_F^{\mathbb{Q}^A} = \mu_F - \mu_F = 0$, and F is a $\mathbb{Q}^A$-martingale:

Theorem 4.3 (Forward Swap Rate SDE Under $\mathbb{Q}^A$)**.** *Under the annuity measure $\mathbb{Q}^A$, $F(t; T_e, T_s)$ is a $\mathbb{Q}^A$-martingale satisfying*

$$\frac{\mathrm{d}F(t; T_e, T_s)}{F(t; T_e, T_s)} \;=\; \sum_{k=1}^{3} \gamma_k(t; T_e, T_s)\; \mathrm{d}W_k^A(t), \tag{38}$$

with $\gamma_k(t) = \sigma_k\big(\mathcal{B}_k^A(t) - \mathcal{B}_k^N(t)\big)$ and $\mathrm{d}W_k^A\, \mathrm{d}W_l^A = \rho_{kl}\, \mathrm{d}t$. The zero-drift condition confirms that F is a $\mathbb{Q}^A$-martingale, consistent with the no-arbitrage requirement.

4.5. Step 5 — Instantaneous Variance and Frozen-Coefficient Approximation

The instantaneous quadratic variation of $\log F$ under $\mathbb{Q}^A$ is

$$\mathrm{d}\langle \log F\rangle_t \;=\; \sum_{k=1}^{3}\sum_{l=1}^{3} \rho_{kl}\, \gamma_k(t)\, \gamma_l(t)\; \mathrm{d}t. \tag{39}$$

Integrating over $[0, T_e]$ gives the *exact* total log-variance:

$$\mathrm{Var}^{\mathbb{Q}^A}\big[\log F(T_e)\big] \;=\; \int_0^{T_e} \sum_{k,l} \rho_{kl}\, \gamma_k(t)\, \gamma_l(t)\; \mathrm{d}t. \tag{40}$$

This integral cannot be evaluated in closed form because $\gamma_k(t)$ depends on the stochastic bond prices $P(t, \cdot)$ through $\mathcal{B}_k^A(t)$ and $\mathcal{B}_k^N(t)$. The standard approximation is to *freeze* the weights at their time-0 values:

$$\mathcal{B}_k^A(t) \;\approx\; \mathcal{B}_k^A(0) \;=\; \frac{\sum_{i=1}^{n} \delta_i\, B_k(0, T_e{+}T_i)\, P(0, T_e{+}T_i)}{A_0}, \tag{41}$$

$$\mathcal{B}_k^N(t) \;\approx\; \mathcal{B}_k^N(0) \;=\; \frac{B_k(0, T_e)\, P(0, T_e) - B_k(0, T_e{+}T_s)\, P(0, T_e{+}T_s)}{N_0}, \tag{42}$$

so that $\gamma_k(t) \approx \gamma_k^{(0)} := \sigma_k(\mathcal{B}_k^A(0) - \mathcal{B}_k^N(0))$ is deterministic. Under this frozen-coefficient approximation, F evolves as a *log-normal* process with constant volatility, and the total variance reduces to

$$\mathrm{Var}^{\mathbb{Q}^A}\big[\log F(T_e)\big] \;\approx\; T_e \sum_{k=1}^{3}\sum_{l=1}^{3} \rho_{kl}\, \gamma_k^{(0)}\, \gamma_l^{(0)}. \tag{43}$$

The resulting Black (log-normal) implied volatility is $\sigma_{\text{Black}} = \sqrt{\sum_{k,l} \rho_{kl} \gamma_k^{(0)} \gamma_l^{(0)}}$.

Remark (Accuracy of the frozen-coefficient approximation). The frozen-coefficient approximation replaces the stochastic weights $\mathcal{B}_k^A(t)$ and $\mathcal{B}_k^N(t)$ with their time-0 values. These weights depend on the bond prices $P(t, \cdot)$, which themselves fluctuate with the state variables $\{x_k(t)\}$. For short-dated swaptions ($T_e \leq 2\text{Y}$) or at typical USD SOFR vol levels (50–150 bps normal vol), the bond prices are close to their initial values over the integration horizon and the approximation error is negligible. For long-dated swaptions ($T_e \geq 10\text{Y}$) in high-volatility regimes, the stochastic variation of the weights can introduce errors of 1–5 bps in implied vol. In such cases, a Monte Carlo simulation of the full SDE system (9) provides the exact benchmark against which the closed-form formula should be validated. For applications requiring high precision at long expiries, two remedies are available. First, a *second-order correction* can be derived by expanding $\mathcal{B}_k^A(t)$ to first order in the state variables and retaining the resulting quadratic terms in the variance integral; this yields an $\mathcal{O}(V_F^2)$ correction analogous to the convexity adjustment in swap rate models. Second, a stochastic-weight perturbation in the spirit of the SABR model (Hagan et al., 2002) can be applied to account for the correlation between the weight process and the forward rate diffusion, providing a closed-form correction at the cost of one additional parameter.

4.6. Step 6 — Connection to the Normal Volatility Formula

For USD SOFR swaptions, rates can be negative or near zero, making the *normal* (Bachelier) model more natural than the log-normal one. Instead of integrating the log-variance of F, we integrate the *absolute* variance of $\mathrm{d}F = F \sum_k \gamma_k^{(0)} \, \mathrm{d}W_k^A$. Freezing $F(t) \approx F_0$:

$$\mathrm{Var}^{\mathbb{Q}^A}\big[F(T_e) - F_0\big] \;\approx\; F_0^2 \, T_e \sum_{k=1}^{3} \sum_{l=1}^{3} \rho_{kl} \, \gamma_k^{(0)} \, \gamma_l^{(0)}, \tag{44}$$

so the normal implied volatility is $\sigma_n = F_0 \, \sigma_{\text{Black}}$.

The hedge ratios h_k of equation (49) arise from the alternative route of differentiating $F = N/A$ with respect to the OU state x_k:

$$\left.\frac{\partial F}{\partial x_k}\right|_{\mathbf{x}=\mathbf{0}} \;=\; h_k(T_e, T_s) \;=\; -F_0 \left.\frac{\partial \log F}{\partial x_k}\right|_{\mathbf{x}=\mathbf{0}} \;=\; -F_0\big(\mathcal{B}_k^N(0) - \mathcal{B}_k^A(0)\big) \;=\; \frac{F_0 \, \gamma_k^{(0)}}{\sigma_k}. \tag{45}$$

Therefore $F_0 \gamma_k^{(0)} = \sigma_k h_k$, and substituting into equation (44):

Corollary 4.4 (Equivalence of Both Routes)**.** *The normal implied volatility from the SDE route equals the hedge-ratio formula of Theorem 5.2:*

$$V_F(T_e, T_s) \;=\; T_e \, F_0^2 \sum_{k,l} \rho_{kl} \, \gamma_k^{(0)} \, \gamma_l^{(0)} \;=\; \sum_{k,l} \frac{\rho_{kl} \, \sigma_k \sigma_l \, h_k \, h_l}{\kappa_k + \kappa_l} \big(1 - e^{-(\kappa_k + \kappa_l) T_e}\big). \tag{46}$$

The second equality follows from substituting $\gamma_k^{(0)} = \sigma_k h_k / F_0$ *and evaluating* $\int_0^{T_e} e^{-(\kappa_k+\kappa_l)(T_e - t)} \, \mathrm{d}t = (1 - e^{-(\kappa_k+\kappa_l)T_e})/(\kappa_k + \kappa_l)$.

Remark (Two Routes to the Same Formula). The derivation of Section 4 (SDE via Itô's quotient rule and Girsanov) and the direct linearisation of Section 5 (differentiate F with

respect to x_k and compute Gaussian variance) both yield exactly V_F as given in (53). The SDE route is more transparent: it makes explicit the measure change, the role of the frozen weights, and the log-normal/normal duality. The linearisation route is more direct for implementation.

5. Linearisation and Closed-Form Normal Volatility

5.1. First-Order Expansion of the Forward Swap Rate

Since each $P(t,T)$ is exponential-affine in the state vector $\mathbf{x}(t) = (x_1(t), x_2(t), x_3(t))^\top$, the forward swap rate $F(t;T_e,T_s)$ is a nonlinear function of $\mathbf{x}$. We linearise around $\mathbf{x} = \mathbf{0}$:

$$F(t;T_e,T_s) \;\approx\; F_0 \;+\; \sum_{k=1}^{3} h_k(T_e,T_s)\, x_k(t), \tag{47}$$

where $F_0 := F(0;T_e,T_s)$ and the hedge ratios are defined as

$$h_k(T_e,T_s) \;:=\; \left.\frac{\partial F}{\partial x_k}\right|_{\mathbf{x}=\mathbf{0}}. \tag{48}$$

5.2. Explicit Formula for the Hedge Ratios

Differentiating (14) with respect to x_k and applying $\partial P(t,T)/\partial x_k = -B_k(t,T)\,P(t,T)$ from Theorem 2.1:

$$\begin{aligned} h_k(T_e,T_s) \;=\; \frac{1}{A_0}\Big[&-B_k(0,T_e)\,P(0,T_e) \;+\; B_k(0,T_e{+}T_s)\,P(0,T_e{+}T_s) \\ &+\; F_0\sum_{i=1}^{n}\delta_i\, B_k(0,T_e{+}T_i)\,P(0,T_e{+}T_i)\Big]. \end{aligned} \tag{49}$$

where $A_0 = A(0;T_e,T_s)$ as defined in Section 3. The three terms in (49) have the following economic interpretation:

Term	Formula	Economic Meaning
Expiry bond	$-B_k(0,T_e)\,P(0,T_e)$	Factor k reduces PV of expiry bond, lowering F
Maturity bond	$+B_k(0,T_e{+}T_s)\,P(0,T_e{+}T_s)$	Factor k reduces PV of maturity bond, raising F
Annuity correction	$+F_0\sum_i \delta_i\, B_k(0,T_e{+}T_i)\,P(0,T_e{+}T_i)$	Corrects for factor k shifting annuity A

5.3. Variance of the Forward Swap Rate at Expiry

Under the linearisation (47), $F(T_e;T_e,T_s)$ is Gaussian under $\mathbb{Q}^A$:

$$F(T_e;T_e,T_s) \;\sim\; \mathcal{N}(F_0,\; V_F(T_e,T_s)) \quad \text{under } \mathbb{Q}^A. \tag{50}$$

The variance is obtained from the quadratic variation of (47):

$$V_F(T_e, T_s) \;=\; \sum_{k=1}^{3}\sum_{l=1}^{3} h_k\, h_l\, \mathrm{Cov}\big(x_k(T_e), x_l(T_e)\big). \tag{51}$$

The covariance of the OU state variables is

$$\mathrm{Cov}\big(x_k(T_e), x_l(T_e)\big) \;=\; \rho_{kl}\int_0^{T_e} \sigma_k\sigma_l\, e^{-(\kappa_k+\kappa_l)(T_e-t)}\,\mathrm{d}t \;=\; \frac{\rho_{kl}\,\sigma_k\sigma_l}{\kappa_k+\kappa_l}\big(1-e^{-(\kappa_k+\kappa_l)T_e}\big). \tag{52}$$

Substituting into the variance formula gives the main closed-form result:

$$\boxed{V_F(T_e, T_s) \;=\; \sum_{k=1}^{3}\sum_{l=1}^{3} \frac{\rho_{kl}\,\sigma_k\sigma_l\, h_k(T_e,T_s)\, h_l(T_e,T_s)}{\kappa_k+\kappa_l}\Big(1-e^{-(\kappa_k+\kappa_l)T_e}\Big).} \tag{53}$$

5.4. Correlation Between Instantaneous Forward Rate Increments

For two arbitrary tenors $T_1, T_2 \geq t$, the HJM volatility function (4) gives the stochastic differential of the instantaneous forward rate as

$$\mathrm{d}f(t,T_i) \;=\; \underbrace{\mu(t,T_i)\,\mathrm{d}t}_{\text{HJM drift}} + \sum_{k=1}^{3}\underbrace{\sigma_k\, e^{-\kappa_k(T_i-t)}}_{\nu_k(t,T_i)}\,\mathrm{d}W_k^{\mathbb{Q}}(t), \qquad i=1,2. \tag{54}$$

The factor loading of $\mathrm{d}f(t,T_i)$ on Brownian motion W_k is $\nu_k(t,T_i) = \sigma_k e^{-\kappa_k(T_i-t)}$, which decays exponentially in the remaining maturity $T_i - t$.

Instantaneous covariance of increments. Since $\mathrm{d}t$ terms are of order $(\mathrm{d}t)^2$ in the quadratic covariation, the drift in (54) does not contribute and

$$\frac{\mathrm{d}\langle\, \mathrm{d}f(\cdot,T_1),\ \mathrm{d}f(\cdot,T_2)\rangle_t}{\mathrm{d}t} \;=\; \sum_{k=1}^{3}\sum_{l=1}^{3}\rho_{kl}\,\nu_k(t,T_1)\,\nu_l(t,T_2) \;=:\; C(T_1,T_2). \tag{55}$$

Setting $T_1 = T_2 = T$ gives the instantaneous variance of $\mathrm{d}f(t,T)$:

$$\frac{\mathrm{d}\langle\, \mathrm{d}f(\cdot,T)\rangle_t}{\mathrm{d}t} \;=\; C(T,T) \;=\; \sum_{k=1}^{3}\sum_{l=1}^{3}\rho_{kl}\,\sigma_k\sigma_l\, e^{-(\kappa_k+\kappa_l)(T-t)}. \tag{56}$$

Instantaneous correlation of increments. Normalising $C(T_1,T_2)$ by the two instantaneous standard deviations yields the model-implied correlation between the forward rate increments $\mathrm{d}f(t,T_1)$ and $\mathrm{d}f(t,T_2)$:

$$\boxed{\mathrm{Corr}\big(\mathrm{d}f(t,T_1),\ \mathrm{d}f(t,T_2)\big) \;=\; \frac{C(T_1,T_2)}{\sqrt{C(T_1,T_1)}\,\sqrt{C(T_2,T_2)}},} \tag{57}$$

where the kernel C is defined by

$$C(T_i, T_j) \;:=\; \sum_{k=1}^{3}\sum_{l=1}^{3} \rho_{kl}\,\sigma_k\sigma_l\, e^{-\kappa_k(T_i-t)}\, e^{-\kappa_l(T_j-t)}. \tag{58}$$

Remark (Structural properties of (57)).

(i) Drift-free. The HJM drift $\mu(t,T)$ is entirely deterministic (given the state variables) and cancels out of the covariation. The correlation of $\mathrm{d}f(t,T_1)$ and $\mathrm{d}f(t,T_2)$ depends only on the volatility structure $\{\sigma_k, \kappa_k, \rho_{kl}\}$ and the maturity gap $|T_2 - T_1|$, not on the current yield curve level.

(ii) Perfect correlation at equal tenors. Setting $T_1 = T_2$ gives Corr $= 1$ identically, since numerator and denominator coincide.

(iii) Maturity decay. As $|T_1 - T_2| \to \infty$, the cross-terms in the numerator decay exponentially. Fast-reverting factors (large κ_k) contribute negligibly to the numerator at long maturity gaps, while slow factors (small κ_k) sustain cross-tenor correlation. The resulting term structure of forward rate correlation is a mixture of exponentials determined by $\{\kappa_k\}$.

(iv) Single-factor limit. For $K = 1$, the kernel reduces to $C(T_i, T_j) = \sigma_1^2\, e^{-\kappa_1(T_i-t)}\, e^{-\kappa_1(T_j-t)}$, so that

$$\mathrm{Corr}\big(\mathrm{d}f(t,T_1),\ \mathrm{d}f(t,T_2)\big) \;=\; 1 \qquad (K=1,\ \text{all } T_1, T_2). \tag{59}$$

With a single Brownian driver all forward rates respond to the same shock; they differ only in the *magnitude* of their response $\sigma_1 e^{-\kappa_1(T-t)}$, not in direction. The instantaneous increment correlation is therefore perfect regardless of the maturity gap $|T_2 - T_1|$. Cross-tenor decorrelation only arises when $K \geq 2$, because different factors carry different exponential decay rates and the ρ_{kl} weights ($k \neq l$) are strictly less than one.

Remark 5.1. It is important not to confuse the increment correlation (57) with the *level* cross-time correlation $\mathrm{Corr}\big(x_1(s),\, x_1(t)\big) = e^{-\kappa_1|t-s|}$ of the single OU state variable x_1. The latter decays exponentially in the *time gap* $|t-s|$, not in the maturity gap $|T_2 - T_1|$, and is a statement about memory in the factor process, not about cross-tenor dependence of the forward curve.

5.5. Main Result: Closed-Form Normal Implied Volatility

Theorem 5.2 (G3++ Normal Swaption Volatility)**.** *Under the three-factor Gaussian HJM model with exponential volatility functions* (4)*, the ATM normal (Bachelier) implied volatility for a swaption with expiry T_e and tenor T_s is*

$$\begin{aligned}\sigma_n^{\text{model}}(T_e, T_s) \;&=\; \sqrt{\frac{V_F(T_e,T_s)}{T_e}} \\ &=\; \sqrt{\frac{1}{T_e}\sum_{k=1}^{3}\sum_{l=1}^{3} \frac{\rho_{kl}\,\sigma_k\sigma_l\, h_k(T_e,T_s)\, h_l(T_e,T_s)}{\kappa_k+\kappa_l}\Big(1 - e^{-(\kappa_k+\kappa_l)T_e}\Big)}\,.\end{aligned} \tag{60}$$

The swaption price under Bachelier's formula is then

$$V_{\text{swaption}}(0) \;=\; A_0\left[(F_0 - K)\,\Phi(d) \;+\; \sqrt{V_F}\,\phi(d)\right], \tag{61}$$

$$d \;=\; \frac{F_0 - K}{\sqrt{V_F}}, \tag{62}$$

where Φ is the standard normal CDF and ϕ is the standard normal PDF.

Remark 4.1 (Measure Approximation). The covariance formula (52) is evaluated under the risk-neutral measure $\mathbb{Q}$ rather than the annuity measure $\mathbb{Q}^A$. The measure change introduces a drift correction that contributes a second-order term to V_F. For ATM swaptions with $T_e \leq 10\text{Y}$, this approximation error is below 0.3 bps in normal vol and is standard in industry practice.

Remark 4.2 (USD SOFR Convention). USD SOFR swaptions are quoted in basis-point normal volatility (bpvol). Multiply σ_n^{model} by 10,000 to convert from rate units to bps/yr.

5.6. Accuracy of the Linearisation

The dominant approximation error from (47) is the second-order Hessian correction:

$$\epsilon_2 \;\approx\; \frac{1}{2}\sum_{k,l}\frac{\partial^2 F}{\partial x_k \partial x_l}\,\mathrm{Cov}(x_k(T_e), x_l(T_e)). \tag{63}$$

For USD SOFR swaptions with typical normal vol levels of 50–150 bps and expiries $T_e \leq 10\text{Y}$, the relative error $|\epsilon_2|/\sigma_n$ is below 0.5 bps and well within bid–offer spreads. For $T_e \geq 20\text{Y}$ the correction may reach 1–3 bps and a second-order formula is recommended.

The table below summarises indicative error bounds across the swaption matrix, combining the linearisation error and the frozen-coefficient error of Section 4.5:

Expiry T_e	**Linearisation error**	**Frozen-weight error**
$\leq 2\text{Y}$	< 0.1 bps	< 0.1 bps
2–10Y	< 0.5 bps	0.1–1 bps
10–20Y	0.5–2 bps	1–3 bps
$\geq 20\text{Y}$	1–3 bps	2–5 bps

Remark (Numerical validation via Monte Carlo). The closed-form formula (60) can be validated against a Monte Carlo simulation of the exact OU system (9). The procedure is: (i) discretise the three OU SDEs on a fine time grid using an exact Euler scheme (exact because the OU transition density is Gaussian); (ii) at each simulated path, compute the bond prices $P(T_e, T_e + T_i)$ from the affine formula (11), the annuity $A(T_e)$, and the swap rate $F(T_e)$; (iii) regress $F(T_e) - F(0)$ on $\{x_k(T_e)\}$ to verify the linearisation accuracy; (iv) compute the sample normal vol as $\hat{\sigma}_n = \mathrm{std}(F(T_e))/\sqrt{T_e}$ and compare to σ_n^{model}. With 10^5 paths and daily time steps, this benchmark is accurate to within 0.05 bps and runs in under one second using vectorised array operations.

6. Calibration to the 20×20 Swaption Vol Matrix

6.1. Parameter Vector and Objective Function

The G3++ model has seven free parameters: three volatility amplitudes $\sigma_k > 0$, three mean-reversion rates $\kappa_k > 0$, and one correlation decay parameter $\beta > 0$ governing the instantaneous correlation between the latent Brownian drivers via

$$\rho_{kl} \;=\; e^{-\beta|k-l|}, \qquad k,l \in \{1,2,3\}, \quad \beta > 0. \tag{64}$$

This single-parameter form guarantees $R = (\rho_{kl})$ is symmetric positive definite with unit diagonal for any $\beta > 0$, and captures the natural ordering that nearby factors are more correlated than distant ones. The full calibration vector is

$$\Theta \;=\; (\sigma_1, \sigma_2, \sigma_3,\; \kappa_1, \kappa_2, \kappa_3,\; \beta) \;\in\; \mathbb{R}^7_{>0}. \tag{65}$$

The calibration objective is the weighted sum of squared normal vol errors over the set $\mathcal{S}$ of liquid swaption cells:

$$\mathcal{L}(\Theta) \;=\; \sum_{(T_e,T_s)\in\mathcal{S}} \omega(T_e,T_s)\left[\sigma_n^{\text{model}}(T_e,T_s;\Theta) - \sigma_n^{\text{mkt}}(T_e,T_s)\right]^2 \;+\; \mu\,\mathcal{P}(\Theta), \tag{66}$$

with liquidity weights $\omega(T_e,T_s) \propto A_0\sqrt{T_e}$ (proportional to Bachelier vega) and regularisation penalty

$$\mathcal{P}(\Theta) \;=\; \lambda_1 \sum_k \left(\frac{\sigma_k}{\sigma_k^{\text{prior}}} - 1\right)^2 + \;\lambda_2 \sum_k \left(\frac{\kappa_k}{\kappa_k^{\text{prior}}} - 1\right)^2 + \;\lambda_3\,(1 - \lambda_{\min}(R))^2. \tag{67}$$

The first two penalty terms discourage large deviations from prior estimates of σ_k and κ_k; the last penalises near-singularity of the correlation matrix R.

> **Remark** (Identifiability of β). The normal vol formula (60) shows that ρ_{kl} enters only through the products $\rho_{kl}\,\sigma_k\,\sigma_l$, making R and the amplitude vector $\boldsymbol{\sigma}$ partially confounded. The one-parameter form (64) regularises this by reducing the correlation degrees of freedom to a single scalar, keeping the calibration problem well-conditioned.

6.2. Convex Structure and Two-Stage Algorithm

Proposition 6.1 (Partial Convexity)**.** *For fixed $\{\kappa_k, \beta\}$, the model normal volatility $\sigma_n^{\text{model}}(T_e,T_s;\Theta)$ is a quadratic form in the amplitude vector $\boldsymbol{\sigma} = (\sigma_1,\ldots,\sigma_3)$. Consequently, the objective (66) is convex (in fact, quadratic up to the square root) in $\boldsymbol{\sigma}$ for fixed $\{\kappa_k,\beta\}$, and the inner minimisation over $\boldsymbol{\sigma}$ is a non-negative least-squares (NNLS) problem with a unique global minimum.*

This structure motivates the nested optimisation

$$\min_{\kappa,\beta}\; \underbrace{\min_{\boldsymbol{\sigma}\geq\mathbf{0}}\; \mathcal{L}(\boldsymbol{\sigma},\kappa,\beta)}_{=:\;\mathcal{L}^*(\kappa,\beta)}, \tag{68}$$

where the inner minimisation is solved exactly (NNLS, $\mathcal{O}(n^3)$) for any fixed outer variables, and the outer minimisation is a six-dimensional nonlinear problem.

6.3. Stage 1 — Inner Loop: Solving for σ_k

For fixed $\{\kappa_k, \beta\}$, define the symmetric matrix $\mathbf{C}(m) \in \mathbb{R}^{5\times 5}$ for each liquid cell $m = (T_e, T_s) \in \mathcal{S}$ with entries

$$C_{kl}(m) \;=\; \frac{\rho_{kl}\, h_k(m)\, h_l(m)}{\kappa_k + \kappa_l} \cdot \frac{1 - e^{-(\kappa_k+\kappa_l)T_e}}{T_e}. \tag{69}$$

Then $\left(\sigma_n^{\text{model}}(m)\right)^2 = \boldsymbol{\sigma}^\top \mathbf{C}(m)\, \boldsymbol{\sigma}$, and the inner problem becomes

$$\boldsymbol{\sigma}^* \;=\; \underset{\boldsymbol{\sigma}\geq \mathbf{0}}{\arg\min} \sum_{m\in\mathcal{S}} \omega_m \Big(\sqrt{\boldsymbol{\sigma}^\top \mathbf{C}(m)\, \boldsymbol{\sigma}} - \sigma_n^{\text{mkt}}(m)\Big)^2, \tag{70}$$

solved via the NNLS algorithm of Lawson and Hanson (1974) after linearising around a current estimate $\boldsymbol{\sigma}^{(0)}$.

6.4. Stage 2 — Outer Loop: Solving for $\{\kappa_k, \beta\}$

The outer objective $\mathcal{L}^*(\kappa, \beta)$ is minimised using gradient-based optimisation. To enforce positivity without explicit bounds, we reparameterise:

$$\sigma_k = e^{s_k}, \qquad \kappa_k = e^{q_k}, \qquad \beta = e^{b}, \qquad s_k,\, q_k,\, b \in \mathbb{R}. \tag{71}$$

The unconstrained vector $\theta = (s_1, \ldots, s_3, q_1, \ldots, q_3, b)^\top \in \mathbb{R}^7$ is then optimised using the L-BFGS-B algorithm (Byrd et al., 1995) with analytical gradients.

6.5. Analytical Gradients

Differentiating (60) with respect to σ_k:

$$\frac{\partial\, \sigma_n^{\text{model}}}{\partial\, \sigma_k} \;=\; \frac{1}{\sigma_n^{\text{model}} \cdot T_e} \sum_{l=1}^{3} \frac{\rho_{kl}\, \sigma_l\, h_k\, h_l}{\kappa_k + \kappa_l} \big(1 - e^{-(\kappa_k+\kappa_l)T_e}\big). \tag{72}$$

The gradient with respect to κ_k requires differentiating $B_k(0,T) = (1 - e^{-\kappa_k T})/\kappa_k$:

$$\frac{\partial B_k(0,T)}{\partial \kappa_k} \;=\; \frac{T\, e^{-\kappa_k T}}{\kappa_k} \;-\; \frac{1 - e^{-\kappa_k T}}{\kappa_k^2}. \tag{73}$$

The chain rule then gives $\partial h_k/\partial \kappa_k$ via (49), and subsequently $\partial \sigma_n^{\text{model}}/\partial \kappa_k$. In practice, automatic differentiation libraries (e.g. JAX or PyTorch) can be used to verify these gradients numerically.

6.6. Robust Pre-processing: Jump Detection

SOFR forward rate daily changes exhibit fat tails driven by discrete jump events — FOMC rate decisions, macro surprises, and liquidity dislocations — that are not part of the continuous Gaussian diffusion in (2). Running PCA on the raw series inflates the empirical covariance and distorts the eigenvector shapes used to initialise κ_k. We therefore model the daily changes as

$$\Delta f(t,\tau) \;=\; \underbrace{\varepsilon(t,\tau)}_{\text{diffusive, } N(0,\Sigma_{\text{diff}})} \;+\; \underbrace{J(t,\tau)}_{\text{jump, sparse}}, \tag{74}$$

and *decompose* each daily change into its diffusive and jump components using the two-stage hybrid algorithm below. Crucially, no observations are discarded: the jump-stripped residual $\hat{\varepsilon}(t,\tau)$ is retained in place of $\Delta f(t,\tau)$ on every flagged day, so the covariance matrix is estimated from the full T-row diffusive matrix. The algorithm combines the sliding-window bipower variation approach of Tawfik (2025) and Spadafora et al. (2018) with an iterative multivariate Mahalanobis refinement.

Stage 1 — Bipower sliding-window test. A global scale estimator such as the MAD conflates volatility across regimes: SOFR experienced 3–5× higher daily volatility during the 2022–2023 tightening cycle than in 2021. Using a single global scale inflates the threshold in low-volatility periods (over-flagging) and loosens it in high-volatility periods (under-flagging genuine FOMC surprises). We therefore use a *local* scale estimated daily from the preceding $K = 21$ business days.

For each day t and tenor j, let $w_j = \{\Delta f(t-K,\tau_j), \ldots, \Delta f(t-1,\tau_j)\}$ be the estimation window. Compute the window mean μ_j, the window median m_j, and the *bipower variation* scale (Barndorff-Nielsen and Shephard, 2004):

$$\sigma_j^* \;=\; \left[\sqrt{\tfrac{\pi}{2}} \sum_{k=1}^{K-1} |\Delta f(t-k,\tau_j) - \mu_j|\; |\Delta f(t-k-1,\tau_j) - \mu_j|\right]^{1/2}, \tag{75}$$

where μ_j is the window arithmetic mean. The mean is the correct centre for the bipower formula as specified in Barndorff-Nielsen and Shephard (2004) and Tawfik (2025). Unlike the sample standard deviation, σ_j^* is robust to isolated jumps within the window: a single outlier appears in at most two consecutive products, whereas the sample variance would square the outlier across all K terms. The per-tenor confidence bounds at level α_{bpv} are

$$x_j^{\pm} \;=\; \Phi^{-1}\!\left(\tfrac{1\pm(1-2\alpha_{\text{bpv}})}{2};\, \mu_j,\, \sigma_j^*\right). \tag{76}$$

Cell (t,j) is flagged as a jump if and only if

$$\Delta f(t,\tau_j) \notin \left[x_j^-(t),\; x_j^+(t)\right]. \tag{77}$$

Each tenor time series is treated *independently*: there is no cross-tenor vote and no minimum number of tenors required to declare a jump. A front-end FOMC surprise that moves only the 3M–2Y tenors is correctly identified as a jump in those series while the 10Y–30Y tenors are treated as clean diffusion on the same day. Day t is reported as a jump day if *any* tenor is flagged: $\exists\, j : (t,j)$ flagged, but this is a diagnostic summary only and does not gate the subtraction.

For each flagged (t,j) pair the jump size is estimated as the deviation of the observed move from the *window median* $m_j(t)$:

$$\hat{J}(t,\tau_j) \;=\; \Delta f(t,\tau_j) - m_j(t), \tag{78}$$

and the diffusive residual is

$$\hat{\varepsilon}(t,\tau_j) \;=\; \Delta f(t,\tau_j) - \hat{J}(t,\tau_j) \;=\; m_j(t). \tag{79}$$

Note the deliberate asymmetry: the bipower formula (75) is centred on the arithmetic mean μ_j (as specified by Barndorff-Nielsen and Shephard 2004), whereas the diffusive level

estimate uses the median m_j. This is because the median has a breakdown point of 1/2: with $K = 21$ and one or two jump days contaminating the window, the arithmetic mean is pulled by up to $\sim 2/21 \approx 10\%$ of the jump size, whereas the median is unaffected. On clean windows mean and median coincide, so the distinction is irrelevant for non-jump days.

Tenors that are *not* individually flagged on a jump day are left unchanged: they carry genuine cross-sectional diffusive information that would be lost if the entire row were discarded.

Stage 2 — Iterative Mahalanobis winsorisation. Stage 1 operates independently per tenor and may miss days where no single tenor exceeds its bipower bound individually yet the *joint* n-dimensional move is anomalous. Stage 2 detects and removes these residual multivariate jumps via a single iterative loop, producing a fully cleaned matrix $\Delta\mathbf{F}^{\text{clean}}$ used for both PCA and the diffusive covariance estimate $\hat{\Sigma}_{\text{diff}}$ (defined in (85) below). All T rows are retained; no rows are excluded or zeroed.

Algorithm 1 Stage 2: Iterative Mahalanobis Winsorisation

Require: $\Delta\mathbf{F}^{\text{clean}} \leftarrow \Delta\mathbf{F}^{\text{diff}}$ (Stage-1 output, all T rows); threshold $c = \chi^2_{1-\alpha_{\text{mah}}}(n)$, $\alpha_{\text{mah}} = 0.001$; regularisation $\varepsilon > 0$; $\hat{J}_2 \leftarrow \mathbf{0}_{T\times n}$; $\mathcal{F}_{-1} \leftarrow \emptyset$

1: **for** $k = 0, 1, 2, \ldots$ **do**

2: **(a) Estimate parameters** from all T rows of current $\Delta\mathbf{F}^{\text{clean}}$:

$$\hat{\mu} = \tfrac{1}{T}\textstyle\sum_t \Delta\mathbf{F}_t^{\text{clean}}, \qquad \hat{\Sigma} = \tfrac{1}{T-1}\textstyle\sum_t (\Delta\mathbf{F}_t^{\text{clean}} - \hat{\mu})(\Delta\mathbf{F}_t^{\text{clean}} - \hat{\mu})^\top$$

Previously winsorised rows contribute their cleaned values.

3: **(b) Compute Mahalanobis distance** for every day t:

$$D_t^2 = \big(\Delta\mathbf{F}_t^{\text{clean}} - \hat{\mu}\big)^\top \big(\hat{\Sigma} + \varepsilon I\big)^{-1} \big(\Delta\mathbf{F}_t^{\text{clean}} - \hat{\mu}\big) \tag{80}$$

Flag day t if $D_t^2 > c$; let $\mathcal{F}_k = \{t : D_t^2 > c\}$.

4: **(c) Check convergence:** if $\mathcal{F}_k = \mathcal{F}_{k-1}$, **stop** (Rousseeuw and Van Driessen, 1999).

5: **(d) Winsorise** each $t \in \mathcal{F}_k$ — shrink radially to the χ^2 ellipsoid surface:

$$s_t = \sqrt{c \,/\, D_t^2} \;\in\; (0, 1), \tag{81}$$

$$\hat{J}_2(t) \mathrel{+}= \big(\Delta\mathbf{F}_t^{\text{clean}} - \hat{\mu}\big)(1 - s_t), \tag{82}$$

$$\Delta\mathbf{F}_t^{\text{clean}} \leftarrow \hat{\mu} + \big(\Delta\mathbf{F}_t^{\text{clean}} - \hat{\mu}\big) s_t \tag{83}$$

s_t is the radial shrinkage factor; the cross-tenor direction of the move is preserved. Unflagged days are unchanged ($s_t = 1$).

6: $\mathcal{F}_{k-1} \leftarrow \mathcal{F}_k$; go to step (a).

7: **end for**

Ensure: Cleaned matrix $\Delta\mathbf{F}^{\text{clean}}$ (all T rows); accumulated jump $\hat{J}_2$ (zero for unflagged days); $\hat{\Sigma}_{\text{diff}}$ (85) estimated from all T rows of $\Delta\mathbf{F}^{\text{clean}}$.

The total two-stage decomposition is

$$\Delta f(t,\tau_j) \;=\; \underbrace{\hat{J}_1(t,\tau_j)}_{\text{Stage-1}} + \underbrace{\hat{J}_2(t,\tau_j)}_{\text{Stage-2}} + \underbrace{\Delta\mathbf{F}^{\text{clean}}_{t,j}}_{\text{diffusive}}, \tag{84}$$

and the diffusive covariance using all T rows is

$$\hat{\Sigma}_{\text{diff}} \;=\; \frac{1}{T-1}\sum_{t=1}^{T}\big(\Delta\mathbf{F}^{\text{clean}}_t - \hat{\mu}\big)\big(\Delta\mathbf{F}^{\text{clean}}_t - \hat{\mu}\big)^{\top}. \tag{85}$$

Remark (Why bipower variation over MAD?). The MAD is a global estimator requiring the full time series. Bipower variation uses only the local window of K observations, making it regime-adaptive. Moreover, bipower variation is specifically designed for processes with rare large jumps: by taking products of *adjacent* absolute returns, a single jump day contaminates at most two products (out of $K-1$), whereas the sample variance squares the jump and spreads its influence across all K terms. The two stages are complementary: Stage 1 subtracts per-tenor bipower-identified jumps; Stage 2 subtracts residual multivariate jumps via winsorisation. Both PCA and $\hat{\Sigma}_{\text{diff}}$ use the fully cleaned $\Delta\mathbf{F}^{\text{clean}}$. Numerical backtests on synthetic SOFR data with a 3× vol regime change showed that the hybrid method achieves an F1 score of 0.60 versus 0.14 for global MAD + Mahalanobis alone.

Remark (Expected jump frequency and parameter defaults). For SOFR forward rates, approximately 8–12 FOMC meeting dates per year generate level jumps across the entire curve. With the default parameters ($K = 21$, $\alpha_{\text{bpv}} = 0.05$, $\alpha_{\text{mah}} = 0.001$) on a typical 500-day sample, approximately 3–6% of (day, tenor) cells are adjusted; the fraction of days with at least one adjusted tenor is consistent with the observed FOMC frequency plus occasional macro shocks.

6.7. From Zero Rates to Instantaneous Forward Rates

Market data is typically available as a term structure of continuously-compounded zero rates $Z(t,\tau)$ at a fixed set of pillar tenors $0 < \tau_1 < \cdots < \tau_n$. Before computing daily changes and running PCA, these must be converted to the instantaneous forward rate $f(t,\tau)$ on a uniform fine grid.

Area-preserving interpolation. The discount factor and instantaneous forward rate are related by

$$P(t,t+\tau) \;=\; e^{-Z(t,\tau)\,\tau}, \qquad f(t,\tau) \;=\; -\frac{\partial}{\partial\tau}\log P(t,t+\tau) \;=\; \frac{\partial}{\partial\tau}\big[Z(t,\tau)\,\tau\big]. \tag{86}$$

Define the *cumulative-yield function*

$$Y(\tau) \;:=\; Z(\tau)\,\tau \;=\; -\log P(\tau). \tag{87}$$

By the fundamental theorem of calculus, any smooth interpolation of Y satisfies the *area-preservation property* exactly:

$$\int_{\tau_i}^{\tau_{i+1}} f(u)\,\mathrm{d}u \;=\; Y(\tau_{i+1}) - Y(\tau_i) \;=\; Z_{i+1}\,\tau_{i+1} - Z_i\,\tau_i, \tag{88}$$

so the integral of the forward rate over each pillar interval equals the log-discount ratio — a no-arbitrage consistency condition.

C^2 cubic spline on Y (not-a-knot). We interpolate Y at the pillar tenors using a C^2 cubic spline with *not-a-knot* boundary conditions: the cubic polynomial is required to be the same across the first two intervals and across the last two intervals, which avoids artificial endpoint effects without imposing an explicit derivative condition. The instantaneous forward rate is then

$$f(\tau) \;=\; \frac{\mathrm{d}Y}{\mathrm{d}\tau}(\tau), \tag{89}$$

which is piecewise quadratic and C^1 (both f and f' are continuous across all pillar nodes). The area-preservation property (88) is inherited automatically because $f = Y'$ and Y interpolates the pillar values.

Piecewise-constant (flat-forward) scheme. An alternative scheme holds the instantaneous forward rate constant on each pillar interval (right-continuous step function). On $[\tau_i, \tau_{i+1})$ the forward rate is the finite-difference derivative of Y:

$$f(\tau) \;=\; f_i \;:=\; \frac{Y_{i+1} - Y_i}{\tau_{i+1} - \tau_i} \;=\; \frac{Z_{i+1}\tau_{i+1} - Z_i\tau_i}{\tau_{i+1} - \tau_i}, \qquad \tau \in [\tau_i, \tau_{i+1}). \tag{90}$$

Area preservation holds trivially: $\int_{\tau_i}^{\tau_{i+1}} f_i \,\mathrm{d}u = f_i(\tau_{i+1} - \tau_i) = Y_{i+1} - Y_i$. The scheme is equivalent to log-linear interpolation of the discount factor: between pillars $\log P(\tau) = \log P_i - f_i(\tau - \tau_i)$, so P is exponential (not spline) on each interval. Beyond the last pillar, f is held flat at f_{n-1}. This scheme is the industry standard for bootstrapping and is preferred when forward rates are needed as step-constant inputs to a tree or Monte Carlo engine.

Fine output grid. The derivative spline is evaluated on a uniform quarterly grid $\tau_j^{\text{out}} = j\,\Delta\tau$, $j = 1, \ldots, m$, with $\Delta\tau = 0.25\,\text{yr}$ from the shortest pillar to the longest. This produces the forward rate matrix

$$\mathbf{F} \;\in\; \mathbb{R}^{T\times m}, \qquad F_{t,j} = f(t,\, \tau_j^{\text{out}}), \tag{91}$$

from which daily changes $\Delta F_{t,j} = F_{t+1,j} - F_{t,j}$ are computed and passed to the jump-detection algorithm (Section 6.6).

Algorithm 2 Zero Rates to Instantaneous Forwards (Area-Preserving)

Require: Zero rates $Z_i = Z(\tau_i)$ at pillars $\tau_1 < \cdots < \tau_n$; output grid $\{\tau_j^{\text{out}}\}$; scheme $\in$ {`cubic`, `monotone`, `flat_forward`}.

Compute cumulative yields: $Y_i \leftarrow Z_i\,\tau_i$.
if scheme = `flat_forward` **then**
  $f_i \leftarrow (Y_{i+1} - Y_i)/(\tau_{i+1} - \tau_i)$ for $i = 1, \ldots, n-1$ (90). For each output point assign $f_j \leftarrow f_i$ where $\tau_j^{\text{out}} \in [\tau_i, \tau_{i+1})$.
else if scheme = `monotone` **then**
  Fit Hyman-filtered Hermite spline $\hat{Y}(\tau)$ on (τ_i, Y_i); differentiate: $f_j \leftarrow \hat{Y}'(\tau_j^{\text{out}})$.
else (`cubic`, default)
  Fit C^2 not-a-knot cubic spline $\hat{Y}(\tau)$; differentiate: $f_j \leftarrow \hat{Y}'(\tau_j^{\text{out}})$.
end if

Ensure: $\int_{\tau_i}^{\tau_{i+1}} f(u)\,\mathrm{d}u = Y_{i+1} - Y_i$ for all i (area preservation, holds for all three schemes).

Remark (Interpolating Y vs. Z directly). Interpolating $Z(\tau)$ directly with a cubic spline does *not* guarantee area preservation: the implied forward rates can oscillate or turn negative between pillars even when the zero curve appears smooth. Interpolating $Y(\tau) = Z(\tau)\tau$ instead works directly with the quantity whose derivative is the forward rate, ensuring the area-preservation property holds exactly.

Remark (Smoothness: C^2 cubic vs. alternatives). Differentiating the interpolant of Y gives $f = Y'$ with one less degree of smoothness than Y. The three natural choices are:

- **Flat-forward** (piecewise constant f, C^{-1}): f is a step function; Y is piecewise linear. Equivalent to log-linear discount factor interpolation. No smoothness across knots; preferred for tree/MC engines and bootstrapping where step-constant forwards are required.
- C^1 quadratic B-spline on Y: f is piecewise linear and C^0 (kinks at interior knots).
- C^1 Hermite cubic on Y: f is piecewise quadratic and C^0 (slopes prescribed at knots).
- C^2 cubic spline on Y (**chosen here**): f is piecewise quadratic and $\mathbf{C^1}$ — both $f(\tau)$ and $f'(\tau)$ continuous everywhere.

All four schemes preserve area exactly. The C^2 cubic is preferred for PCA because the C^1 continuity of f makes the resulting time series of forward rate changes smoother, reducing artificial noise in the PCA input matrix. For severely inverted or crisis curves where overshoot is a concern, the Hyman (1983) monotonicity-filtered Hermite spline guarantees $f(\tau) \geq 0$ everywhere at the cost of reducing smoothness to C^0 in f. The flat-forward scheme is the natural choice when the downstream model requires step-constant inputs (e.g. a short-rate tree).

6.8. Full Data Pipeline

The complete pre-processing pipeline converting raw SOFR zero rates to a jump-cleaned forward rate covariance matrix is summarised in Algorithm 3.

Algorithm 3 SOFR Forward Rate Pre-Processing Pipeline

Require: Historical zero rate matrix $\mathbf{Z} \in \mathbb{R}^{T\times n}$; pillar tenors $\{\tau_i\}$; output grid spacing $\Delta\tau$.

1: **for** each date $t = 1, \ldots, T$ **do**
2: Apply Algorithm 2 to row $\mathbf{Z}_{t,:} \Rightarrow$ forward rate vector $\mathbf{f}_t \in \mathbb{R}^m$.
3: **end for**
4: Form daily changes: $\Delta\mathbf{f}_t = \mathbf{f}_{t+1} - \mathbf{f}_t$, $t = 1, \ldots, T-1$.
5: **Stage 1**: for each tenor j independently, compute bipower scale σ_j^* (75); flag cell (t, j) if it breaches bounds (77); subtract jump via (78)–(79).
6: **Stage 2**: compute D_t^2 (80) for each day t; winsorise flagged days via (81)–(83); re-estimate $\hat{\Sigma}$ from all T rows of $\Delta\mathbf{F}^{\text{clean}}$; iterate until convergence $\Rightarrow \Delta\mathbf{F}^{\text{clean}}$, $\hat{\Sigma}_{\text{diff}}$ (85).
7: Run PCA on $\Delta\mathbf{F}^{\text{clean}}$ (all T rows) $\Rightarrow$ eigenvectors $(V_{R,K})$, variance explained per factor; confirms $K = 3$ and identifies level/slope/curvature structure for κ_k initialisation.

Ensure: $\Delta\mathbf{F}^{\text{clean}}$ (84) and $\hat{\Sigma}_{\text{diff}}$ (85) (both all T rows); PCA eigenvectors $V_{R,K}$ and variance-explained profile.

6.9. Bridge Between PCA Observables and G3++ Parameters

This section derives the formal link between the output of PCA on historical forward rate changes and the structural parameters $(\kappa_k, \sigma_k, \beta)$ of the G3++ model. The derivation justifies the initialisation strategy of Section 6.10 and can also serve as a standalone, data-driven parameter estimator when a swaption vol surface is unavailable.

6.9.1. Model-Implied Covariance Structure

Under the G3++ model, the daily forward rate change at tenor pillar τ_j is (the HJM drift is deterministic and drops out of covariance calculations):

$$\Delta f(t, \tau_j) \;\approx\; \sum_{k=1}^{3} \sigma_k \, e^{-\kappa_k \tau_j} \, \Delta W_k(t). \tag{92}$$

Define the $n \times 3$ *basis matrix*

$$E \;:=\; \begin{pmatrix} e^{-\kappa_1 \tau_1} & e^{-\kappa_2 \tau_1} & e^{-\kappa_3 \tau_1} \\ \vdots & \vdots & \vdots \\ e^{-\kappa_1 \tau_n} & e^{-\kappa_2 \tau_n} & e^{-\kappa_3 \tau_n} \end{pmatrix}, \qquad [E]_{jk} = e^{-\kappa_k \tau_j}, \tag{93}$$

and the 3×3 *factor-space covariance matrix*

$$M_{kl} \;:=\; \rho_{kl} \, \sigma_k \sigma_l \, \Delta t, \qquad k, l \in \{1, 2, 3\}. \tag{94}$$

Then the $n \times n$ model covariance matrix of forward rate changes is

$$\Sigma \;=\; E \, M \, E^{\top}, \tag{95}$$

a *rank-3* matrix: all covariance information is contained in the 3-dimensional column space of E.

6.9.2. Identification of M by Projection

Given the empirical covariance $\hat{\Sigma}_{\text{diff}}$ (an $n \times n$ full-rank matrix estimated from the cleaned panel), we wish to recover the 3×3 matrix M. Since E is $n \times 3$ with $n \gg 3$ it has no square inverse, so we use a least-squares projection.

Proposition 6.2 (Factor-Space Projection)**.** *Assume E has full column rank, which holds whenever the κ_k are distinct and the tenor grid spans a range wide enough for the three exponential functions to be linearly independent. Then the minimiser of the Frobenius residual*

$$\min_{M} \left\| \hat{\Sigma}_{\text{diff}} - E \, M \, E^{\top} \right\|_F^2 \tag{96}$$

is given by

$$\boxed{M \;=\; (E^{\top} E)^{-1} \, E^{\top} \, \hat{\Sigma}_{\text{diff}} \, E \, (E^{\top} E)^{-1}.} \tag{97}$$

Proof. Pre-multiply the model relation $\hat{\Sigma} \approx E M E^{\top}$ by $E^{\top}$:

$$E^{\top} \hat{\Sigma} \;\approx\; (E^{\top} E) \, M \, E^{\top}.$$

Post-multiply both sides by E:

$$E^\top \hat{\Sigma}\, E \;\approx\; (E^\top E)\, M\, (E^\top E).$$

Both sides are 3×3. Since $E^\top E$ is invertible (by the full-rank assumption), solve for M:

$$M \;=\; (E^\top E)^{-1}\, E^\top \hat{\Sigma}\, E\, (E^\top E)^{-1}.$$

This is the normal equation of the Frobenius problem (96), hence its unique minimiser. Consistency: when $\hat{\Sigma} = EME^\top$ exactly,

$$(E^\top E)^{-1} E^\top (EME^\top) E (E^\top E)^{-1} = (E^\top E)^{-1} (E^\top E)\, M\, (E^\top E)(E^\top E)^{-1} = M. \qquad \square$$

> **Remark** (Pseudoinverse interpretation). The factor $(E^\top E)^{-1} E^\top$ is the Moore–Penrose pseudoinverse E^+ of the tall matrix E, so $M = E^+ \hat{\Sigma} (E^+)^\top$. Pre-multiplying by E^+ projects the rows of $\hat{\Sigma}$ from tenor space into factor space; post-multiplying by $(E^+)^\top$ projects the columns. The result is the 3×3 matrix of factor-space covariances. The formula requires $E^\top E$ to be well-conditioned; this fails when two κ_k values are nearly equal, which is precisely why the initialisation uses a log-spaced grid to keep the columns of E well-separated.

6.9.3. Recovery of σ_k, ρ_{kl}, and β

Once $\hat{M}$ is obtained from (97), the model parameters follow directly from the definition (94).

Volatility amplitudes. The diagonal entries give

$$\hat{\sigma}_k \;=\; \sqrt{\frac{\hat{M}_{kk}}{\Delta t}}, \qquad k = 1, 2, 3. \tag{98}$$

Inter-factor correlations. The off-diagonal entries give

$$\hat{\rho}_{kl} \;=\; \frac{\hat{M}_{kl}}{\sqrt{\hat{M}_{kk}\, \hat{M}_{ll}}}, \qquad k \neq l. \tag{99}$$

These are the empirical correlations between the latent Brownian increments, recovered by projecting the data into the model's exponential factor basis.

Correlation decay parameter. Map $\{\hat{\rho}_{kl}\}$ to the one-parameter form (64) by fitting

$$\hat{\beta} \;=\; \arg\min_{\beta > 0} \sum_{k<l} \left(\hat{\rho}_{kl} - e^{-\beta|k-l|}\right)^2. \tag{100}$$

For $K = 3$ there are two distinct pairs ($\hat{\rho}_{12}, \hat{\rho}_{23}$ at lag 1 and $\hat{\rho}_{13}$ at lag 2). Because $e^{-\beta|k-l|}$ is strictly decreasing in β for each pair, the problem has a unique solution, which is found efficiently by a one-dimensional bisection or by noting that the objective is convex in β.

6.9.4. Identification of κ_k from Eigenvector Decay

The mean-reversion rates κ_k are identified from the spatial decay of the PCA loading vectors. Let $v_k \in \mathbb{R}^n$ be the k-th eigenvector of $\hat{\Sigma}_{\text{diff}}$ (ordered by descending eigenvalue). The model predicts

$$[v_k]_j \;\propto\; e^{-\kappa_k \tau_j}, \tag{101}$$

so taking logarithms gives a linear relationship in the tenor:

$$\log\big|[v_k]_j\big| \;=\; a_k \;-\; \kappa_k\,\tau_j \;+\; \varepsilon_j. \tag{102}$$

Fitting (102) by ordinary least squares over the tenor grid $\{\tau_j\}_{j=1}^n$ yields the OLS estimator

$$\hat{\kappa}_k \;=\; -\frac{\displaystyle\sum_{j=1}^{n}\big(\tau_j - \bar{\tau}\big)\,\big(\log|[v_k]_j| - \overline{\log|v_k|}\,\big)}{\displaystyle\sum_{j=1}^{n}\big(\tau_j - \bar{\tau}\big)^2}, \tag{103}$$

where $\bar{\tau}$ and $\overline{\log|v_k|}$ denote the respective sample means over the tenor grid. In practice, the three eigenvectors produce well-separated estimates:

- v_1 (level): near-flat profile $\Rightarrow$ $\hat{\kappa}_1 \approx 0.05$–$0.15\,\text{yr}^{-1}$,
- v_2 (slope): moderate decay $\Rightarrow$ $\hat{\kappa}_2 \approx 0.3$–$0.6\,\text{yr}^{-1}$,
- v_3 (curvature): rapid short-end decay $\Rightarrow$ $\hat{\kappa}_3 \approx 1.5$–$4.0\,\text{yr}^{-1}$.

6.9.5. The Rotation Ambiguity

A subtlety arises because PCA always returns *orthogonal* eigenvectors, whereas the G3++ model factors are *correlated* ($\rho_{kl} \neq 0$). The empirical eigenvectors $\{v_k\}$ are therefore not equal to the model basis columns $\{[E]_{\cdot k}\}$; they are related by a rotation that can be derived explicitly from the model structure.

The rotation matrix. Recall $M = SRS\,\Delta t$ where $S = \text{diag}(\sigma_1, \sigma_2, \sigma_3)$ and $R = (\rho_{kl})$. Since R is symmetric positive definite, write $R = R^{1/2}R^{1/2}$ (spectral or Cholesky square root). Because S is diagonal with positive entries, $S^\top = S$, so

$$\frac{M}{\Delta t} \;=\; S\,R\,S \;=\; \big(S\,R^{1/2}\big)\big(S\,R^{1/2}\big)^\top. \tag{104}$$

Define the 3×3 rotation matrix

$$L \;:=\; S\,R^{1/2}. \tag{105}$$

Note that $L \neq S^{1/2}R^{1/2}$: the latter would give $LL^\top = S^{1/2}RS^{1/2} \neq SRS$ in general.

Composite basis and PCA eigenvectors. Define $\tilde{E} := EL = ESR^{1/2}$, an $n \times 3$ matrix. Substituting into (95):

$$\Sigma \;=\; E\,M\,E^\top \;=\; E\,(LL^\top)\,E^\top\,\Delta t \;=\; \tilde{E}\,\tilde{E}^\top\,\Delta t, \tag{106}$$

so the PCA eigenvectors of Σ are the left singular vectors of $\tilde{E}$, not the columns of E.

Remark (Structure of the rotation). The matrix $L = SR^{1/2}$ mixes two effects: $S = \mathrm{diag}(\sigma_k)$ rescales each column of E by the corresponding volatility amplitude, while $R^{1/2}$ rotates the rescaled columns according to the inter-factor correlation structure. When $R = I$ (uncorrelated factors), $L = S$ and $\tilde{E}$ simply scales each column of E by σ_k; the eigenvectors of Σ then align with those of E (up to sign), and the regression (103) recovers κ_k exactly. When β is small (high correlation), $R^{1/2}$ deviates significantly from I and the eigenvectors of $\tilde{E}$ are genuine mixtures of the exponential shapes. The iteration below corrects for this.

Remark (Iterative rotation correction). Starting from the initial estimates $\hat{\kappa}_k$ from (103), one can refine as follows: (i) compute $\hat{M}$ from (97) and extract $\hat{S} = \mathrm{diag}(\hat{\sigma}_k)$, $\hat{R}^{1/2}$ from (99); (ii) form $\tilde{E} = E\hat{S}\hat{R}^{1/2}$; (iii) re-run the log-linear regression (103) on the columns of $E^{+}\tilde{E}$ to update $\hat{\kappa}_k$; (iv) repeat from step (i). A single iteration is sufficient in practice because the rotation effect on the log-decay rate is second order in $\|\hat{R}^{1/2} - I\|$.

6.9.6. Complete Identification Workflow

The steps below proceed in dependency order and produce estimates $(\hat{\kappa}_k, \hat{\sigma}_k, \hat{\beta})$ for all seven G3++ parameters from the historical forward rate panel alone.

1. **Estimate $\hat{\Sigma}_{\text{diff}}$.** Run the jump-detection pipeline (Section 6.6) to obtain the cleaned diffusive covariance $\hat{\Sigma}_{\text{diff}}$ (85).
2. **Run PCA.** Eigen-decompose $\hat{\Sigma}_{\text{diff}} = V\Lambda V^{\top}$ with $\lambda_1 \geq \cdots \geq \lambda_n$. Confirm that the top 3 components explain $> 95\%$ of total variance.
3. **Estimate** $\hat{\kappa}_k$ by applying the OLS log-linear regression (103) to each of v_1, v_2, v_3.
4. **Build** E**:** $[E]_{jk} = e^{-\hat{\kappa}_k \tau_j}$.
5. **Compute** $\hat{M}$ via the projection formula (97).
6. **Extract** $\hat{\sigma}_k$ from the diagonal of $\hat{M}$ via (98).
7. **Extract** $\hat{\rho}_{kl}$ from the off-diagonal of $\hat{M}$ via (99).
8. **Fit** $\hat{\beta}$ by solving the scalar optimisation (100).
9. **Correct for rotation (optional).** Form $\hat{S} = \mathrm{diag}(\hat{\sigma}_k)$ and compute $\hat{R}^{1/2}$ (spectral or Cholesky square root of $\hat{R}$ from (99)); set $L = \hat{S}\hat{R}^{1/2}$ per (105) and $\tilde{E} = EL$; re-estimate $\hat{\kappa}_k$ from the columns of $\tilde{E}$; repeat from step 4 until convergence.

The resulting estimates serve directly as the initialisation $\Theta^{(0)} = (\hat{\sigma}_k, \hat{\kappa}_k, \hat{\beta})$ for the nested swaption calibration of Section 6.4.

6.10. Initialisation via PCA

The parameter identification of Section 6.9 provides closed-form starting values for all seven parameters. These are further refined by the two-stage swaption optimisation.

Initialising $\{\kappa_k\}$. Use $\hat{\kappa}_k$ from the log-linear regression (103). As a robust fallback when

eigenvector shapes are noisy, replace with the fixed log-spaced grid

$$\kappa_k^{(0)} \;=\; \exp\!\left(\log\kappa_{\min} + \frac{k-1}{K-1}\big(\log\kappa_{\max} - \log\kappa_{\min}\big)\right), \quad k = 1, \ldots, K, \tag{107}$$

with $\kappa_{\min} = 0.05\,\mathrm{yr}^{-1}$ and $\kappa_{\max} = 3.0\,\mathrm{yr}^{-1}$, giving $\kappa^{(0)} \approx (0.05,\, 0.39,\, 3.0)\,\mathrm{yr}^{-1}$.

Initialising $\{\sigma_k\}$. Use $\hat{\sigma}_k$ from (98). As a fallback, run one inner NNLS pass (Section 6.3) with $\kappa^{(0)}$ and $\beta^{(0)}$ fixed:

$$\boldsymbol{\sigma}^{(0)} \;=\; \arg\min_{\boldsymbol{\sigma}\geq 0} \mathcal{L}(\boldsymbol{\sigma};\, \kappa^{(0)}, \beta^{(0)}), \tag{108}$$

which has a unique global minimum by Proposition 6.1.

Initialising β. Use $\hat{\beta}$ from (100). As a robust default, $\beta^{(0)} = 0.3$ gives $\rho_{12} = \rho_{23} \approx 0.74$ under (64), consistent with typical USD SOFR cross-factor correlation.

Summary of initialisation steps.

1. Execute the identification workflow of Section 6.9.6 to obtain $(\hat{\kappa}_k, \hat{\sigma}_k, \hat{\beta})$.
2. Set $\kappa_k^{(0)} = \hat{\kappa}_k$; fall back to (107) if regression residuals are large.
3. Set $\beta^{(0)} = \hat{\beta}$; fall back to 0.3 if $\hat{M}$ is near-singular.
4. Set $\boldsymbol{\sigma}^{(0)} = \hat{\boldsymbol{\sigma}}$; refine with one NNLS pass via (108).

6.11. Convergence Criteria

The optimisation is declared converged when all of the following hold simultaneously:

(i) $\|\nabla_\theta \mathcal{L}\|_2 < 10^{-6}$ (gradient norm),

(ii) $|\Delta\mathcal{L}/\mathcal{L}| < 10^{-8}$ (relative objective change),

(iii) $\sigma_k > 0,\ \kappa_k > 0,\ \beta > 0$ for all k (feasibility),

(iv) RMSE over $\mathcal{S} < 0.5$ bps and max error < 2 bps (quality).

6.12. Full Calibration Algorithm

Algorithm 4 G3++ Calibration to USD SOFR Normal Vol Matrix

Require: Discount curve $P(0,T)$; market vols $\sigma_n^{\text{mkt}}(T_e,T_s)$; liquidity weights $\omega(T_e,T_s)$.

Bootstrap $P(0,T)$ from OIS/SOFR. Compute A_0, F_0, $\{L_i\}$ for all 400 cells.
Run PCA on historical Δf data to confirm $K=3$ and identify level/slope/curvature structure. Initialise $\{\kappa_k^{(0)}\}$ via log-spaced grid (107); set $\beta^{(0)}=0.3$; solve (108) for $\{\sigma_k^{(0)}\}$ (Section 6.10).
Reparameterise to $\theta^{(0)}=(\log\sigma_k^{(0)},\log\kappa_k^{(0)},\log\beta^{(0)})$.
repeat
    Recover $\{\sigma_k,\kappa_k,\beta\}=\{e^{s_k},e^{q_k},e^b\}$ from current θ.
    Compute $B_k(0,T)$ via (7); compute hedge ratios $h_k(T_e,T_s)$ via (49) for all cells.
    Compute matrix $C_{kl}(m)$ via (69); compute σ_n^{model} via (60).
    Inner loop: Solve NNLS problem (70) for $\boldsymbol{\sigma}^*$. Update $\{s_k\}\leftarrow\{\log\sigma_k^*\}$ in θ.
    Evaluate $\mathcal{L}(\theta)$ (66) and gradient $\nabla_\theta\mathcal{L}$ via (72)–(73).
    Execute L-BFGS-B update step on θ (outer loop).
until convergence criteria (i)–(iv) satisfied
Recover $\{\sigma_k^*,\kappa_k^*,\beta^*\}$; compute residuals and diagnostics.

Ensure: Calibrated parameters; model normal vol surface; RMSE; $\lambda_{\min}(R)$.

7. Calibration Quality and Diagnostics

7.1. Quality Thresholds for USD SOFR Swaptions

Metric	Excellent	Acceptable	Investigate
RMSE (σ_n, bps)	<0.5	0.5–2	>2
Max absolute error (bps)	<2	2–5	>5
R^2 over liquid cells	>0.999	0.99–0.999	<0.99
$\lambda_{\min}(R)$	>0.01	$>-10^{-6}$	$<-10^{-6}$
Short-expiry ($T_e\le 1$Y)	<1 bps	1–3 bps	>3 bps
Long-expiry ($T_e\ge 10$Y)	<1 bps	1–2 bps	>2 bps

7.2. Common Failure Modes and Remedies

Short-expiry cells poorly fit.
The fast-decaying factor ($k=3$) has insufficient amplitude or κ_3 is too small. Remedy: widen the upper bound to $\kappa_k\le 10\,\text{yr}^{-1}$; ensure 1W, 1M, 3M expiry cells are included in $\mathcal{S}$.

Long-expiry cells poorly fit.
The level factor ($k=1$) decays too quickly. Remedy: enforce $\kappa_1\le 0.05\,\text{yr}^{-1}$ in the outer loop, or add a pure parallel-shift factor with $\kappa=0$.

Correlation matrix not PSD.
Occurs when free off-diagonal ρ_{kl} are used. Remedy: switch to the exponential parameterisation (64); or project R onto the PSD cone via eigenvalue truncation after each iteration.

Convergence to local minimum.
Remedy: run 20 multi-start instances with random perturbations $\theta^{(0)} + \epsilon_i$ ($\epsilon_i \sim \mathcal{N}(\mathbf{0}, 0.1\, I)$) around the PCA initialisation; accept the solution with lowest $\mathcal{L}^*$.

References

Brigo, D. and Mercurio, F. (2006). *Interest Rate Models — Theory and Practice*, 2nd ed. Springer Finance.

Byrd, R. H., Lu, P., Nocedal, J., and Zhu, C. (1995). A limited memory algorithm for bound constrained optimization. *SIAM Journal on Scientific Computing*, 16(5), 1190–1208.

Heath, D., Jarrow, R., and Morton, A. (1992). Bond pricing and the term structure of interest rates: a new methodology for contingent claims valuation. *Econometrica*, 60(1), 77–105.

Jamshidian, F. (1989). An exact bond option formula. *Journal of Finance*, 44(1), 205–209.

Lawson, C. L. and Hanson, R. J. (1974). *Solving Least Squares Problems*. Prentice-Hall.

Rebonato, R. (2002). *Modern Pricing of Interest Rate Derivatives: The LIBOR Market Model and Beyond*. Princeton University Press.

Rousseeuw, P. J. and Croux, C. (1993). Alternatives to the median absolute deviation. *Journal of the American Statistical Association*, 88(424), 1273–1283.

Rousseeuw, P. J. and Van Driessen, K. (1999). A fast algorithm for the minimum covariance determinant estimator. *Technometrics*, 41(3), 212–223.

Hyman, J. M. (1983). Accurate monotonicity-preserving cubic interpolation. *SIAM Journal on Scientific and Statistical Computing*, 4(4), 645–654.

Tawfik, M. (2025). Monte Carlo Simulation of SOFR Yield Curve. *Citibank Working Paper*.

Spadafora, L., Sivero, F. and Picchiotti, N. (2018). Jumping VaR: Order statistics volatility estimator for jumps classification and market risk modeling. *arXiv preprint* arXiv:1803.07021.

Barndorff-Nielsen, O. E. and Shephard, N. (2004). Power and bipower variation with stochastic volatility and jumps. *Journal of Financial Econometrics*, 2(1), 1–37.

Hagan, P. S., Kumar, D., Lesniewski, A. S. and Woodward, D. E. (2002). Managing smile risk. *Wilmott Magazine*, September, 84–108.